

TITLE SLIDE | FINAL PRESENTATION FILE UPLOADED

IBM Data Analyst Capstone Project

Project title: IBM Data Analyst Capstone Project

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Uploaded file: DataAnalystPresentation.pdf

This PDF is the complete final presentation file. It contains the Title, Outline, Executive Summary, Introduction, Methodology, Programming Languages Trends, Findings and Implications, Database Trends, Dashboard Tab 1, Dashboard Tab 2, Dashboard Tab 3, Discussion, Overall Findings and Implications, and Conclusion slides.

OUTLINE SLIDE

- Title Slide (project title, learner's name, date)
- Outline Slide
- Executive Summary Slide (key findings, bullet points)
- Introduction Slide (purpose, target audience, value)
- Methodology Slide (sources, collection, wrangling)
- Programming Languages Trends (current year and next year bar charts)
- Programming Languages Trends - Findings and Implications

- Database Trends (currently in use and future demand)
- Database Trends - Findings and Implications
- Dashboard Slide with three tabs and charts
- Dashboard Tab 1 - Current Technology Usage
- Dashboard Tab 2 - Future Technology Trends
- Dashboard Tab 3 - Demographics
- Discussion Slide (insights from the dashboard)

EXECUTIVE SUMMARY SLIDE | CONCLUSION SLIDE

Executive Summary - key findings

- JavaScript 62.3%, HTML/CSS 52.9%, Python 51.0%, SQL 51.0% are most used; Python is most desired next year (41.9%).
- PostgreSQL leads databases in use (48.7%) and future demand (47.1%).
- Job postings peak in Washington DC (5,316), Detroit, Seattle and New York.
- Recommend Python, SQL and PostgreSQL; monitor Rust and Go.

Conclusion - project conclusions

- The project analyzed the Stack Overflow survey, job-postings and salary data.
- Python, JavaScript, SQL and PostgreSQL are the strongest all-round skills.
- Rust, Go and SQLite are high-growth technologies worth investment.
- These conclusions inform hiring, curriculum design and technology roadmaps.

INTRODUCTION SLIDE | DISCUSSION SLIDE

Introduction

- Purpose of the report: identify in-demand technology skills from the Stack Overflow Developer Survey, job-postings.xlsx and salary data.
- Target audience: hiring managers, curriculum designers, and job seekers.
- Overall value: a data-driven view of current versus future technology adoption for recruiting and upskilling.

Discussion - dashboard insights

- Dashboard Tab 1 Current Technology Usage: JavaScript, Python, PostgreSQL, AWS and React/Node.js dominate current use.
- Dashboard Tab 2 Future Technology Trends: Python and PostgreSQL lead desire; Rust, Go and SQLite are rising.
- Dashboard Tab 3 Demographics: full-time developers aged 25-34, mainly United States, Germany and India.
- Key insights derived from the dashboard visualizations: usage and future desire are linked but not identical.

METHODOLOGY SLIDE | OVERALL FINDINGS AND IMPLICATIONS SLIDE

Methodology

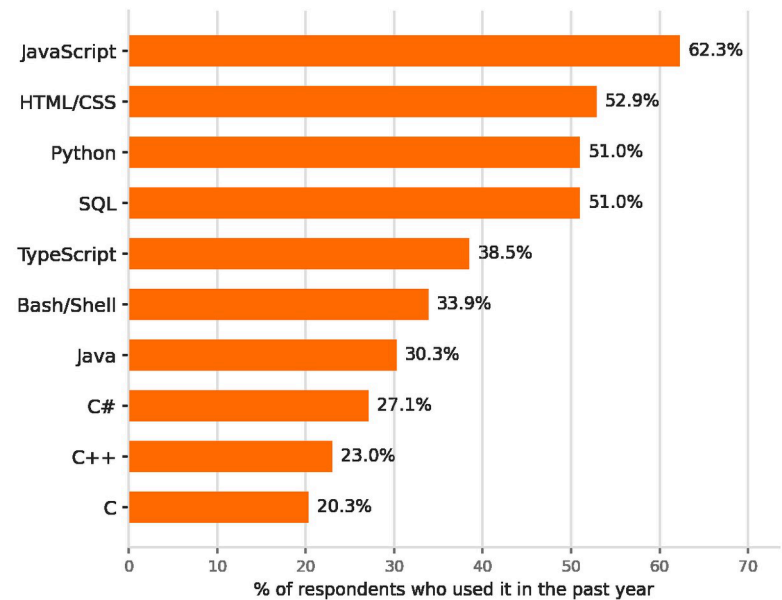
- Data sources: Stack Overflow Developer Survey 2024; job-postings.xlsx (Jobs API); popular-languages.csv (web scraping).
- Data collection methods: pandas CSV load, requests API, BeautifulSoup HTML parsing.
- Key data wrangling steps: remove duplicates; standardize labels; impute missing values; Min-Max and Z-score; IQR outlier removal.

Overall findings and industry implications

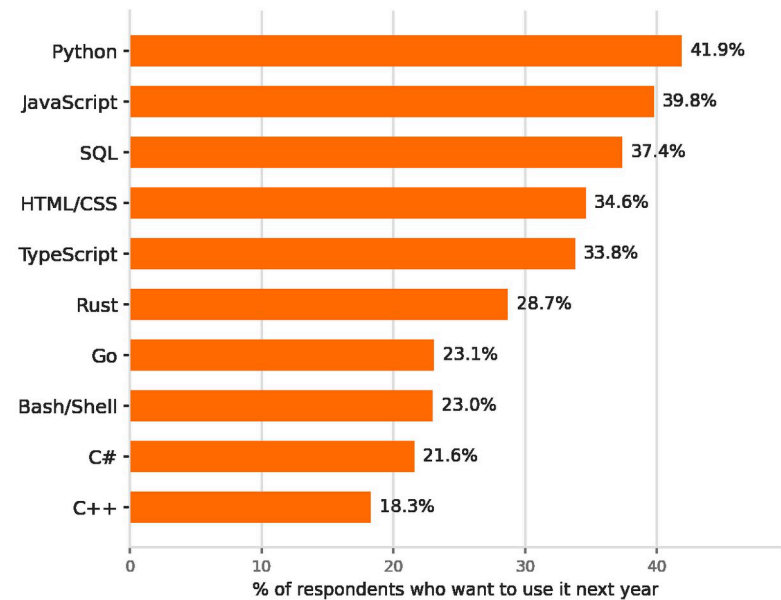
- Significant results: Python and PostgreSQL win on usage, desire and jobs; Rust, Go and SQLite are emerging.
- Real-world / industry implications: treat Python and PostgreSQL as baseline skills; add Rust, Go and SQLite electives; keep compensation location-aware.
- These results connect survey sentiment to hiring and curriculum decisions.

PROGRAMMING LANGUAGES TRENDS | FINDINGS AND IMPLICATIONS

Bar chart: top 10 current year



Bar chart: top 10 next year



Findings and implications

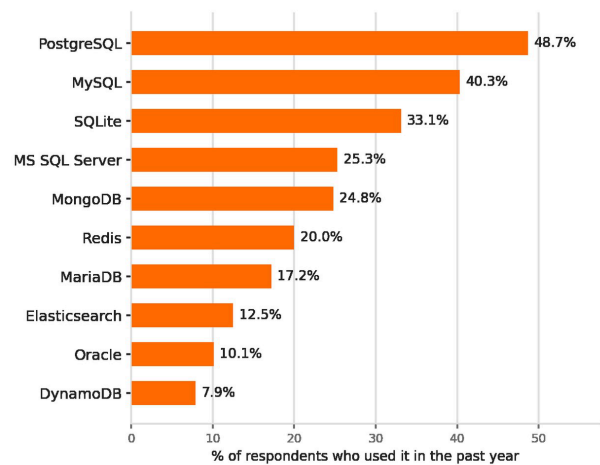
- Findings: JavaScript, HTML/CSS and Python lead current use; Python overtakes JavaScript next year (41.9% vs 39.8%); Rust and Go have the strongest momentum.
- Implications: make Python a core hiring skill; keep JavaScript/TypeScript; pilot Rust and Go.

Programming Languages Trends - Findings and Implications: overall findings based on the programming languages chart are that Python becomes the most-desired language while JavaScript remains widely used. Implications for industry are to require Python, retain JavaScript, and explore Rust and Go.

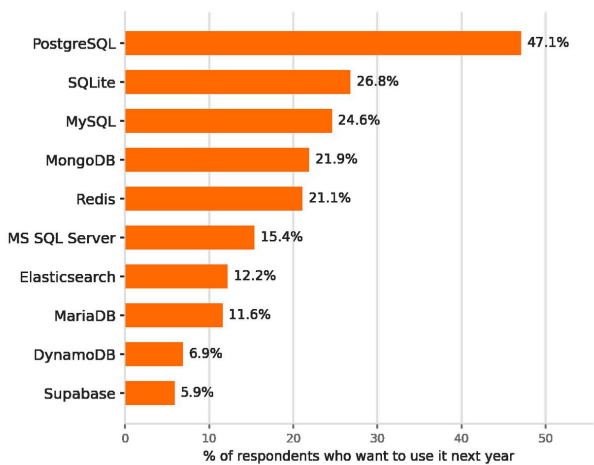
Database Trends currently in use and future demand plus findings. Dashboard Tab 1 Current Technology Usage, Tab 2 Future Technology Trends, Tab 3 Demographics with charts.

DATABASE TRENDS | FINDINGS AND IMPLICATIONS | DASHBOARD TABS 1-3

Currently in use

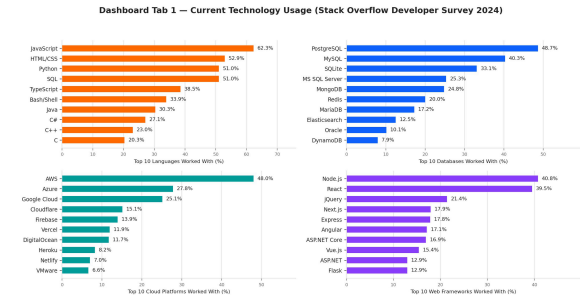


Anticipated future demand

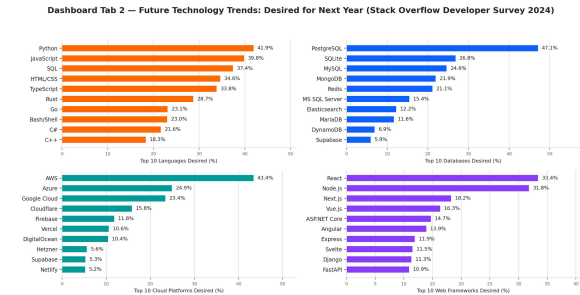


Database Trends findings and implications: PostgreSQL leads current use (48.7%) and future demand (47.1%); SQLite overtakes MySQL. Implications: make PostgreSQL the default; teach SQLite; keep MongoDB and Redis.

Dashboard Tab 1 Current Technology Usage



Dashboard Tab 2 Future Technology Trends



Dashboard Tab 3 Demographics

